Ahmedabad University

ECE 501: Digital Image Processing

Week 7 Progress Report

Group 6

Vehicle Orientation Estimation Using Conventional Image Processing

Submitted to faculty: Prof. Mehul Raval

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GitHub Repository (DRASHTI-HaOBB):

https://github.com/ERYAGNIK003/DRASHTI-Dataset

Dataset Source:

DRASHTI-HaOBB – Heading-angle Oriented Bounding Box Dataset

https://zenodo.org/records/18278989

1. Work Done This Week

In Week 7, we focused on comparing the performance of the previously implemented vehicle orientation-estimation methods and investigating the reasons behind their angular errors. During the earlier weeks, three conventional image-processing methods, namely PCA, Hough Transform, and contour-based estimation, were implemented. In Week 6, the Hough Transform was further modified by introducing short-line filtering and dominant orientation grouping.

The main objective of Week 7 was to understand the limitations of these methods using the results obtained from the existing 100 vehicle observations. Instead of introducing another orientation-estimation algorithm, we concentrated on analyzing the errors produced by the existing methods.

First, we loaded the original Week 3 results containing the vehicle identifiers, classes, ground-truth orientations, and estimated orientations from PCA, Hough Transform, and contour-based estimation. The reconstructed vehicle crops saved during Week 6 were also reused for evaluating the Week 6 baseline and modified Hough methods.

The same 100 vehicle observations were retained throughout the analysis. All 100 corresponding vehicle crops were successfully located, and no additional vehicle samples were introduced.

Next, we calculated the overall angular errors for the five available prediction outputs: PCA, historical Week 3 Hough, contour-based estimation, Week 6 baseline Hough, and Week 6 modified Hough. These results were compared using mean error, median error, maximum error, standard deviation, and the number of high-error predictions.

A class-wise analysis was then performed to examine whether certain vehicle categories produced larger orientation-estimation errors. A heatmap was generated to compare the mean angular errors of the different methods across the vehicle classes.

We also investigated whether multiple methods produced high errors on the same vehicle observations. For this analysis, an angular error of 30° or greater was considered a high-error prediction. The vehicles were grouped according to the number of methods that produced high errors.

Another part of the work involved comparing the Week 6 baseline Hough method with the modified Hough method on all 100 observations. The individual angular-error differences were calculated to identify the vehicles for which the modification improved or worsened the estimated orientation.

Finally, selected vehicle observations were visualized using their original crops, Canny edge maps, all detected Hough line segments, and the line segments selected by the modified Hough method. These visualizations were generated to investigate the possible reasons for inaccurate orientation estimation.

The Week 7 workflow was:

Week 3 Results CSV + Week 6 Vehicle Crops → Loading and Validation → Re-evaluation of Week 6 Hough Methods → Overall Error Comparison → Class-wise Analysis → Shared High-Error Analysis → Improvement and Regression Analysis → Visual Analysis → Saving Graphs and Tables

2. Methodology

2.1 Loading the Previous Results

The Week 3 results CSV file was loaded using the Pandas library. It contained the vehicle identifiers, vehicle classes, ground-truth orientations, and previously estimated orientations from the three original methods.

The saved predictions of PCA, Original Hough Transform, and contour-based estimation were reused rather than implementing these methods again.

This allowed us to retain the earlier experimental results and compare them with the newly evaluated Hough methods.

The ground-truth and predicted orientations were represented in the 0°–90° range, following the orientation convention used in the previous experiments.

2.2 Reusing the Vehicle Crops

The vehicle crops reconstructed during Week 6 were reused for the Week 7 analysis.

The vehicle crop manifest was loaded, and the crop paths were checked against the original 100 vehicle identifiers.

Each crop was checked to confirm that the corresponding image could be loaded successfully.

The following checks were performed:

1. Load the original 100 vehicle identifiers from the Week 3 results.
2. Read the Week 6 vehicle crop manifest.
3. Match the vehicle identifiers with the saved crop paths.
4. Check whether each crop file exists and can be loaded.
5. Retain all original vehicle observations for the analysis.

All 100 vehicle crops were available. Therefore, no vehicle sample had to be removed or replaced during the Week 7 evaluation.

2.3 Re-evaluation of the Hough Transform Methods

The Week 6 baseline and modified Hough methods were evaluated again using the reconstructed vehicle crops.

The same general preprocessing pipeline from Week 6 was used. It consisted of:

- Grayscale conversion.
- Contrast enhancement using CLAHE.
- Gaussian smoothing.
- Canny edge detection.
- Morphological closing.

The preprocessing settings included a CLAHE clip limit of 2.0, an 8 × 8 tile grid, a 5 × 5 Gaussian kernel, Canny thresholds of 50 and 150, and a 3 × 3 morphological closing kernel.

Baseline Hough Transform

The Probabilistic Hough Transform was applied to the edge image to detect line segments.

The orientation and length of each detected line segment were calculated. The line lengths were used as weights while combining the detected orientations through a length-weighted axial mean.

The baseline parameters were:

Parameter	Value
Hough threshold	30
Minimum line length	20 pixels
Maximum line gap	10 pixels

The resulting axis was converted into the required 0°–90° orientation representation.

Modified Hough Transform

The modified Hough method implemented during Week 6 was reused without further parameter tuning.

The modification involved two main operations:

1. **Short-line filtering:** Detected line segments shorter than a specified fraction of the crop diagonal were removed.
2. **Dominant orientation selection:** The remaining line segments were grouped according to their orientations, and the group with the greatest combined line length was selected.

A length-weighted axial mean was then calculated using the selected line segments.

The selected parameters from Week 6 were:

Parameter	Selected value
Hough threshold	30
Minimum line length	15 pixels
Maximum line gap	5 pixels
Minimum relative line length	0.05 × crop diagonal
Orientation bin width	20°

The Week 6 parameter configuration was reused for all 100 vehicle observations. No additional parameter tuning was performed during Week 7.

2.4 Angular Error Calculation

The estimated orientations from the different methods were compared with the corresponding ground-truth orientations.

The angular error was calculated as:

$$Angular\ Error = |\theta_{predicted} - \theta_{ground\ truth}|$$

Both angles followed the project's 0°–90° orientation convention.

The following performance measures were calculated:

- Number of valid predictions.
- Number of unavailable predictions.
- Mean angular error.
- Median angular error.
- Maximum angular error.
- Standard deviation of angular error.
- Number of predictions with an angular error of 30° or greater.

The mean angular error was used to compare the overall performance of the methods, while the median and maximum errors were considered to understand the distribution of individual prediction errors.

2.5 Vehicle-Class-Wise Analysis

The 100 vehicle observations were grouped according to their vehicle classes.

For every class and method, the following values were calculated:

- Number of vehicle observations.
- Number of valid predictions.

- Mean angular error.
- Median angular error.
- Maximum angular error.
- Number of high-error predictions.

A heatmap was generated to visualize the class-wise mean angular errors across the different methods.

This analysis was used to identify vehicle categories that produced comparatively high errors and to examine whether similar difficulties were observed across multiple methods.

The number of observations in each class was also considered while interpreting the results because the original 100-image sample was not equally distributed among the vehicle categories.

2.6 Identification of Common High-Error Observations

To investigate whether multiple methods produced large errors on the same vehicle, a high-error threshold of 30° was selected.

For each vehicle, the number of prediction outputs with an angular error greater than or equal to 30° was counted.

The shared-failure analysis considered four prediction outputs:

1. PCA.
2. Historical Week 3 Hough Transform.
3. Contour-based estimation.
4. Modified Hough Transform.

The recomputed Week 6 baseline was excluded from this particular count because the purpose was to compare the original method outputs and the modified method.

The vehicles were then grouped into the following categories:

- No high-error predictions.
- One method producing a high error.
- Two methods producing high errors.
- Three methods producing high errors.
- All four methods producing high errors.

A table and graph were generated to represent the distribution of these observations.

2.7 Modified Hough Improvement and Regression Analysis

The Week 6 baseline and modified Hough methods were compared on the same 100 vehicle observations.

For each vehicle, the difference between the two angular errors was calculated as:

$$Error\ Difference\ =\ Modified\ Hough\ Error\ -\ Week\ 6\ Baseline\ Error$$

The difference was interpreted as follows:

- Negative difference: Modified Hough produced a lower angular error.
- Positive difference: Modified Hough produced a higher angular error.
- Zero difference: Both methods produced the same angular error.

The individual results were sorted to identify the vehicles with the largest improvements and regressions.

This analysis helped examine whether the modifications were beneficial for all vehicle observations or whether some vehicles were affected negatively by the line filtering and dominant orientation selection operations.

2.8 Visual Analysis of Difficult Vehicle Observations

Numerical angular-error values indicate which vehicle predictions were inaccurate, but they do not independently establish the reasons behind those errors.

Therefore, selected vehicle observations were examined using visualizations generated from the modified Hough method.

The selection included:

- Vehicles with the highest Modified Hough angular errors.
- Vehicles for which Modified Hough performed worse than the Week 6 baseline.
- Vehicles where multiple orientation-estimation outputs produced high errors.

After removing duplicate vehicle identifiers, 19 unique vehicle observations were selected for visual analysis.

For each selected observation, four visualizations were combined:

1. Original vehicle crop.
2. Canny edge map.
3. All detected Hough line segments.
4. Line segments selected by Modified Hough.

These images were used to examine whether the detected lines represented the expected longitudinal direction of the vehicle.

The visual analysis mainly supported the analysis of Hough-based estimation. The original PCA and contour-based predictions were included in the numerical comparison, but separate PCA-axis and contour-rectangle overlays were not generated during this stage.

2.9 Saving the Results

The generated graphs, tables, and vehicle visualizations were saved in the Week7_Results folder.

The saved outputs included:

- Overall comparison table.
- Complete 100-vehicle comparison table.
- Class-wise comparison table.
- Shared-failure tables.

- Modified Hough improvement table.
- Modified Hough regression table.
- Manual failure-analysis table.
- Class-wise error heatmap.
- Shared high-error count graph.
- Individual vehicle visualizations.

This organisation allows the results to be reviewed and reused during the remaining stages of the project.

3. Results and Observations

3.1 Reuse and Validation of the Vehicle Crops

The Week 7 workflow successfully loaded all 100 original vehicle observations from the Week 3 results.

The Week 6 crop manifest was used to locate the corresponding vehicle crops.

All 100 vehicle crops were successfully located and loaded. No unavailable crops were found during this stage.

The same vehicle identifiers were retained throughout the analysis, ensuring consistency with the earlier experiments.

3.2 Overall Comparison of Orientation-Estimation Methods

The overall performance of the five available prediction outputs was calculated using the same 100 vehicle observations.

The results are presented below.

Method	Valid Predictions	Unavailable Predictions	Mean Error	Median Error	Maximum Error	Standard Deviation	High Errors (≥30°)
PCA	100	0	10.74°	4.85°	81.71°	16.11°	8
Original Hough (Week 3)	98	2	12.46°	4.18°	82.39°	20.44°	11
Contour-based	100	0	10.97°	5.72°	89.00°	16.44°	12
Week 6 Baseline Hough	100	0	7.85°	3.98°	82.68°	14.54°	5

Modified Hough	100	0	**5.35°**	**2.02°**	84.94°	11.12°	**3**

Table 1. Overall comparison of orientation-estimation methods on the 100-vehicle sample.

The results show that Modified Hough produced the lowest mean angular error of 5.35° among the evaluated methods.

The Week 6 baseline Hough method produced a mean angular error of 7.85°, while the modified method reduced it to 5.35° on the complete 100-vehicle sample.

The median error also decreased from 3.98° for the Week 6 baseline to 2.02° for Modified Hough.

The number of high-error predictions decreased from 5 for the Week 6 baseline to 3 for Modified Hough.

However, the maximum error increased from 82.68° to 84.94°. This indicates that although the modified method reduced the overall mean and the number of high-error predictions, some individual vehicles continued to produce very large errors.

The historical Week 3 Hough method produced a mean angular error of 12.46° with two unavailable predictions. These historical results are retained as a reference because the original Week 3 implementation and the reconstructed Week 6 preprocessing could not be confirmed to be identical.

The main comparison for evaluating the Week 6 modification is therefore between the Week 6 baseline and modified Hough methods.

It should also be noted that the 100-vehicle Modified Hough evaluation includes the 70 observations used for selecting its parameters during Week 6. Therefore, this result is useful for exploratory comparison and error investigation, but it should not replace the independent held-out evaluation reported in Week 6.

The Week 6 held-out experiment had shown a baseline mean error of 8.01° and a modified mean error of 8.81°. The difference between these results and the complete 100-vehicle comparison is related to the different evaluation subsets and should be considered while interpreting the performance.

3.3 Vehicle-Class-Wise Error Analysis

The class-wise analysis was performed to examine whether the methods produced different levels of error for different vehicle categories.

The mean angular errors for the five prediction outputs are presented below.

Vehicle Class	Samples	PCA	Original Hough	Contour	Week 6 Baseline	Modified Hough
Auto3WCargo	3	11.34°	38.58°	13.41°	14.09°	5.06°
AutoRicksaw	8	7.49°	13.97°	8.29°	18.00°	4.77°
Bus	2	27.32°	0.73°	24.18°	2.11°	0.75°

Container	3	40.12°	49.13°*	27.49°	2.76°	0.59°
MotorCycle	30	7.28°	4.54°*	10.15°	4.23°	6.13°
PickUp	4	8.73°	20.72°	3.86°	6.69°	2.05°
SUV	34	7.64°	12.97°	8.34°	10.00°	5.80°
Sedan	7	23.55°	7.03°	18.10°	8.47°	1.78°
Tanker	1	0.19°	3.27°	7.00°	4.00°	1.29°
Trailer	3	28.97°	32.87°	30.21°	0.73°	0.42°
Truck	5	8.05°	17.50°	6.39°	5.53°	14.91°

Table 2. Class-wise mean angular error comparison across the 100 vehicle observations.

(Note: Original Hough produced two unavailable predictions in the 100-vehicle sample. The Container mean is based on two valid predictions out of three, and the MotorCycle mean is based on 29 valid predictions out of 30.)

The class-wise results show that the performance of the methods varies across vehicle categories.

For PCA, Container showed the highest mean angular error at 40.12°, followed by Trailer at 28.97° and Bus at 27.32°.

For the historical Original Hough method, Container had the highest mean error at 49.13°, followed by Auto3WCargo at 38.58° and Trailer at 32.87°.

For contour-based estimation, Trailer had the highest mean error at 30.21°, followed by Container at 27.49° and Bus at 24.18°.

The Modified Hough method produced a lower mean error for most of the represented vehicle classes. However, Truck showed the highest Modified Hough mean error at 14.91°, based on five samples. MotorCycle and SUV had mean errors of 6.13° and 5.80°, respectively.

The results also show that some classes produced comparatively high errors across more than one method. Container, Trailer, Bus, and Sedan are examples where PCA and contour-based estimation showed relatively high mean errors.

However, some vehicle classes contain very few observations. For example, Tanker has only one sample, while Bus has two and Container and Trailer have three each. Therefore, these class-wise values should be treated as observations from the current sample rather than final conclusions about the entire vehicle categories.

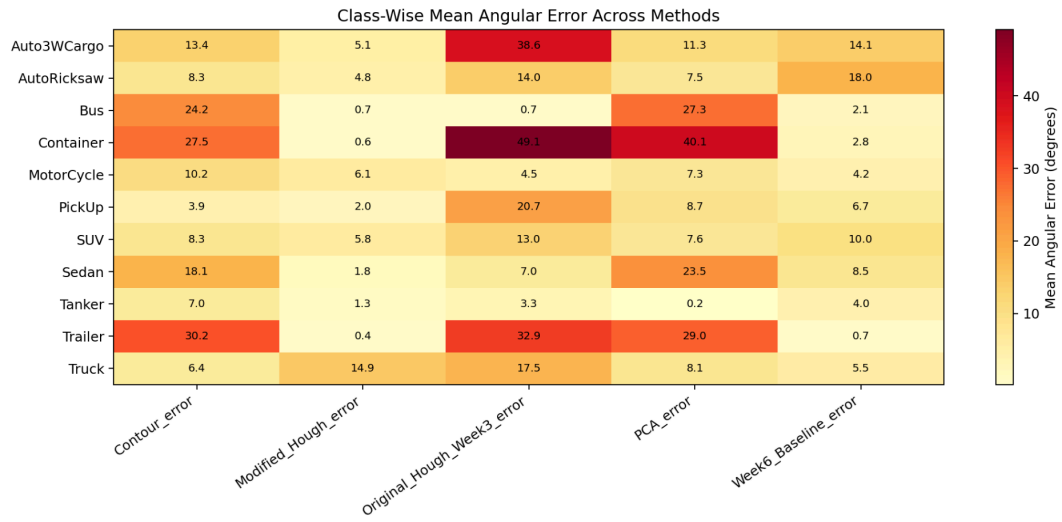


Figure 1. Heatmap showing class-wise mean angular errors across PCA, Original Hough, Contour-Based, Week 6 Baseline Hough, and Modified Hough methods.

3.4 Common High-Error Vehicle Analysis

The shared high-error analysis was performed using a threshold of 30°.

The results showed the following distribution:

Number of Methods with Error ≥30°	Number of Vehicles
0	75
1	18
2	5
3	2
4	0
Total	**100**

Table 3. Distribution of vehicles according to the number of methods producing high errors.

The analysis showed that 75 vehicles had no recorded prediction error of 30° or greater among the available outputs. Unavailable predictions were not counted as high-error predictions.

Eighteen vehicles had a high error in only one method, while five vehicles had high errors in two methods.

Two vehicles had high errors in three methods. No vehicle had a high error across all four prediction outputs.

Therefore, seven vehicle observations showed high errors in at least two methods.

The two observations with high errors in three methods were:

- cLEH_vQvJ_zAUO_vehicle_1 – Trailer.
- y0bo_oHBY_y2mz_vehicle_1 – Container.

The results indicate that some vehicle observations are difficult for multiple conventional image-processing methods. However, a shared high error does not automatically confirm that the methods failed because of the same reason.

The seven shared high-error observations are listed below.

Vehicle ID	Vehicle Class	Number of High-Error Outputs	Methods with Error ≥30°
WB91_YzXz_82r4_vehicle_1	Container	2	PCA, Contour
tc9H_Ng2A_DWcC_vehicle_1	Bus	2	PCA, Contour
Kg67_1U8u_Rx2q_vehicle_1	MotorCycle	2	PCA, Contour
dDHH_wv1m_jsew_vehicle_1	SUV	2	PCA, Contour
hn8n_2cU8_01No_vehicle_1	Sedan	2	PCA, Contour
y0bo_oHBY_y2mz_vehicle_1	Container	3	PCA, Original Hough, Contour
cLEH_vQvJ_zAUO_vehicle_1	Trailer	3	PCA, Original Hough, Contour

Table 4. Vehicle observations with high errors in at least two prediction outputs.

The shared high-error analysis shows that PCA and contour-based estimation produced high errors on several of the same vehicles. This suggests that some observations may contain shape or boundary characteristics that are difficult for more than one estimation method.

For the Container and Trailer examples, the original Hough method also produced high errors. However, Modified Hough produced much lower errors on these same observations.

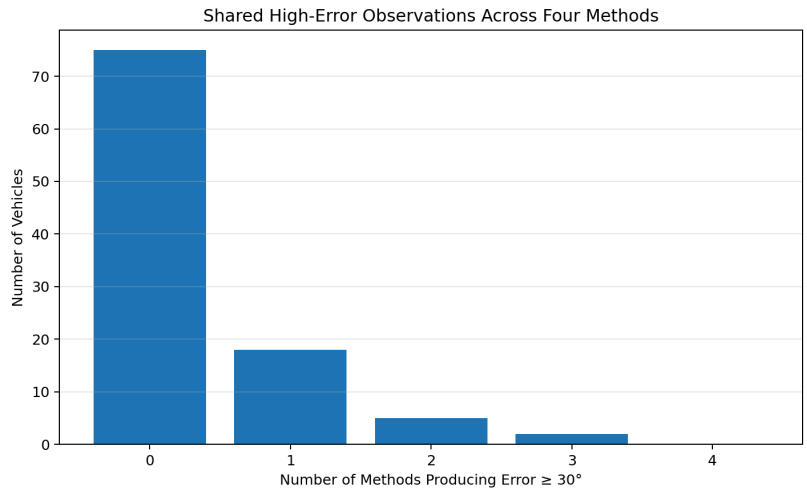


Figure 2. Number of vehicle observations grouped according to the number of methods producing an angular error of 30° or greater.

3.5 Modified Hough Improvement and Regression Analysis

The Week 6 baseline and modified Hough methods were compared for all 100 vehicle observations.

The individual comparison showed that:

Result	Number of Vehicles
Improved predictions	65
Worsened predictions	35
Unchanged predictions	0
Total evaluated vehicles	**100**

Table 5. Summary of Modified Hough improvements and regressions on the 100-vehicle sample.

Modified Hough reduced the angular error for 65 vehicles and increased the error for the remaining 35 vehicles.

Although the modified method worsened the predictions for 35 vehicles, the improvements in other observations resulted in a lower overall mean error.

The mean angular error decreased from 7.85° for the Week 6 baseline to 5.35° for Modified Hough, corresponding to an average reduction of approximately 2.50° on this complete 100-vehicle sample.

The largest improvements and regressions were also identified.

Vehicle ID	Vehicle Class	Week 6 Baseline Error	Modified Hough Error	Error Reduction
gqbe_NCQI_MTLw_vehicle_1	SUV	78.96°	1.35°	77.61°
uKNC_Y7V6_crcn_vehicle_1	AutoRicksaw	81.04°	6.72°	74.32°
xqcd_XrBY_xvxc_vehicle_1	SUV	44.52°	0.00°	44.52°

Table 6. Examples of the largest improvements obtained using Modified Hough.

Vehicle ID	Vehicle Class	Week 6 Baseline Error	Modified Hough Error	Increase in Error
lPCV_CAG5_X9at_vehicle_2	Truck	14.16°	59.82°	45.66°
5QtK_2GHz_wRDq_vehicle_1	SUV	6.85°	34.00°	27.15°
K8fi_E7BV_E33r_vehicle_1	MotorCycle	3.27°	23.67°	20.40°

Table 7. Examples of the largest regressions observed with Modified Hough.

The results show that Modified Hough improved many vehicle predictions, but it also produced large regressions for certain observations.

For example, the error for lPCV_CAG5_X9at_vehicle_2, which belongs to the Truck class, increased from 14.16° to 59.82°. Similarly, the error for 5QtK_2GHz_wRDq_vehicle_1, an SUV, increased from 6.85° to 34.00°.

These cases indicate that the short-line filtering and dominant orientation selection operations may not be suitable for every vehicle image.

The modification can improve the estimate when the selected line group represents the vehicle's main axis. However, it can also increase the error when useful line segments are removed or a different orientation group is selected.

The visual analysis was therefore used to examine these cases more closely.

3.6 Visual Analysis of Difficult Vehicle Observations

A total of 19 unique vehicle observations were selected for visual analysis.

The selection included the ten vehicles with the highest Modified Hough errors, the ten largest regressions compared with the Week 6 baseline, and the vehicles having high errors across multiple methods. Duplicate observations were removed before generating the final visualizations.

For each selected observation, four images were combined:

1. Original vehicle crop.
2. Canny edge map.
3. All detected Hough line segments.
4. Modified Hough selected line segments.

The purpose of these visualizations was to examine the relationship between the actual vehicle appearance, detected edges, and the line segments used for estimating orientation.

The following examples were selected for the report.



Figure 3. Hough line analysis of the Truck sample where Modified Hough produced a large regression.



Figure 4. Hough line analysis of the SUV sample where Modified Hough produced a large regression.

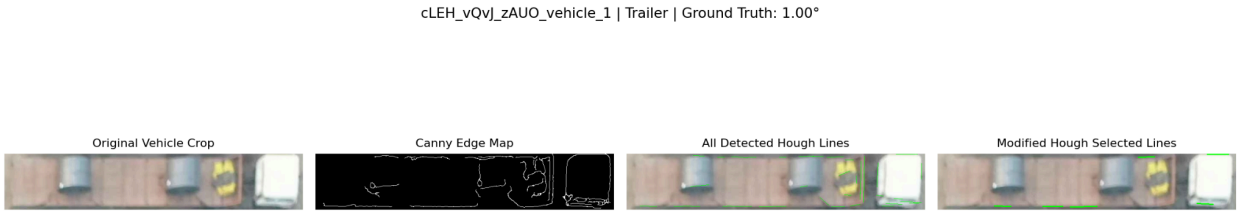


Figure 5. Hough line analysis of the Trailer sample with high errors in PCA, Original Hough, and contour-based estimation.

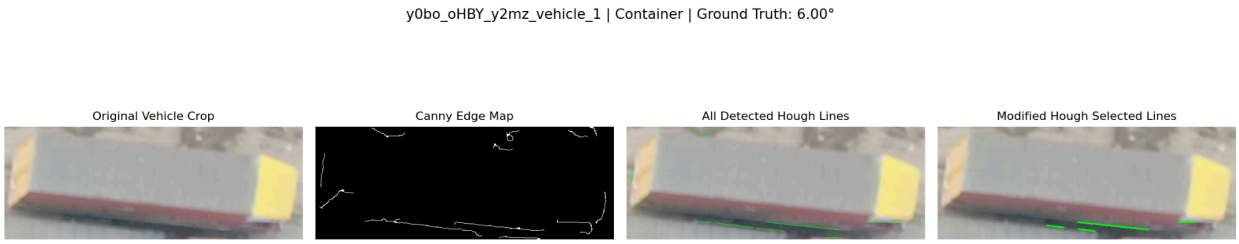


Figure 6. Hough line analysis of the Container sample with high errors in PCA, Original Hough, and contour-based estimation.

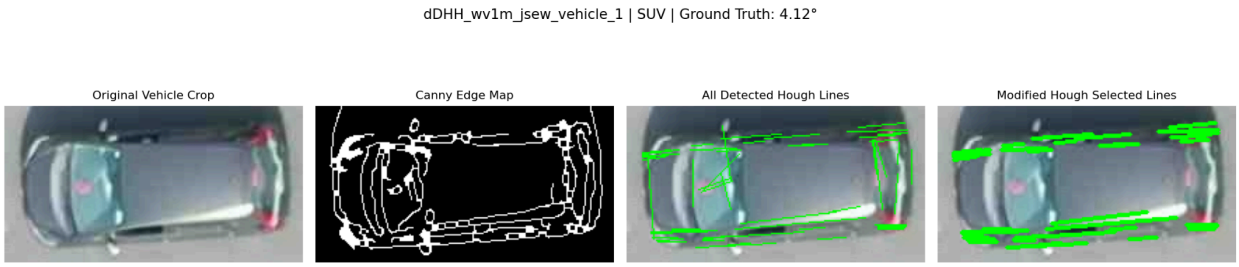


Figure 7. Hough line analysis of the SUV sample with high errors in PCA and contour-based estimation.

The visualizations provide a way to inspect the line structures detected by the Hough Transform and the orientation group selected by the modified method.

They also help identify cases where the edge map contains line structures in several directions rather than one clearly dominant direction.

The visual analysis provides diagnostic information about Hough line detection. However, the specific reason behind each high-error prediction needs to be confirmed through individual inspection rather than being inferred only from the numerical error.

3.7 Investigation of Possible Error Sources

The Week 7 analysis helped identify several possible factors that may contribute to inaccurate vehicle orientation estimation.

1. Vehicle Geometry

Different vehicle classes have different shapes and structural arrangements. Some vehicles may not have a clearly distinguishable longitudinal direction from the top-view image.

For example, vehicles with rectangular bodies may contain strong edges along both their length and width. This can make it difficult for an orientation-estimation method to identify the correct principal direction.

2. Background Interference

The vehicle crop may contain surrounding structures, road markings, shadows, or other edges.

These additional structures may influence edge detection and line detection, particularly when the vehicle boundaries are not clearly separated from the background.

3. Weak or Fragmented Boundaries

Some vehicle boundaries may not appear as continuous edges because of image quality, low resolution, or limited contrast.

This can affect contour extraction and the line segments detected by the Hough Transform.

4. Incorrect Dominant-Line Selection

The Modified Hough method selects the orientation group with the greatest combined line length.

However, the group containing the greatest line length does not always represent the actual longitudinal direction of the vehicle. Strong lines from windows, roofs, or other structures may influence the selected direction.

5. Removal of Useful Line Segments

The short-line filtering operation removes line segments below a selected relative length threshold.

Although this can reduce the influence of short or irrelevant lines, it may also remove useful edge information in some vehicle images.

6. Image Quality and Vehicle Appearance

Low-resolution, blurred, or unclear vehicle crops may affect the reliability of the detected contours and line segments.

These factors can influence the performance of conventional image-processing methods, especially when the vehicle shape does not provide strong and continuous boundaries.

The numerical results and visualizations provide a starting point for investigating these limitations. However, the above factors are treated as possible explanations and are not considered confirmed causes for every individual high-error prediction.

The PCA and contour-based predictions were reused from Week 3, and separate PCA-axis and contour-rectangle overlays were not generated in Week 7. Therefore, the exact internal reasons for their high errors require further visual analysis.

3.8 Observations and Limitations of the Analysis

The Week 7 analysis provided a comparison of the orientation-estimation methods on the original 100 vehicle observations.

The results show that Modified Hough achieved a lower mean angular error than the Week 6 baseline on the complete 100-vehicle sample. However, the earlier held-out evaluation showed a different result, with the modified method producing a slightly higher mean error than the baseline.

This difference is important because the parameters of Modified Hough were selected using 70 of the original 100 observations during Week 6. Therefore, the complete 100-vehicle results include observations that were used during parameter selection.

The class-wise analysis was also affected by the uneven distribution of vehicle categories. Some classes contained many observations, while others contained only one or a few samples.

The Week 7 analysis retained the same 100 observations and did not include a separate experiment with different sample sizes. Therefore, it does not establish that increasing the number of images directly causes the mean angular error to increase.

The analysis instead provides information about how errors are distributed across the existing observations and vehicle categories.

4. Challenges Faced

One of the main challenges during Week 7 was comparing the original Week 3 predictions with the Hough results recalculated during Week 6. The original PCA, Hough, and contour-based predictions were reused, whereas the Week 6 baseline and modified Hough methods were evaluated again using the reconstructed vehicle crops.

Because the original Week 3 crop construction and preprocessing could not be confirmed to be exactly identical to the Week 6 implementation, the historical and recomputed results had to be treated as separate references.

Another challenge was interpreting the class-wise error results. Some vehicle categories had very few observations, making their mean angular errors sensitive to individual predictions.

Identifying shared high-error observations was also important because different methods could produce large errors on the same vehicle. However, a shared high error does not necessarily mean that the methods failed for the same reason.

The visual analysis presented another challenge. The generated Hough visualizations showed the detected and selected line segments, but numerical comparisons alone could not confirm the exact reason behind every incorrect prediction.

Finally, the modified Hough method improved the errors for many observations but also produced large regressions for certain vehicles. Understanding why the selected line group changed the estimated orientation was therefore an important part of the analysis.

5. Current Status

The Week 7 work extended the previous orientation-estimation experiments by conducting a comparative error analysis and investigating the limitations of the existing methods.

The work completed includes:

- Reusing the same 100 vehicle observations from the earlier experiments.
- Loading and validating all 100 reconstructed vehicle crops.
- Reusing the historical PCA, Original Hough, and contour-based predictions.
- Re-evaluating the Week 6 baseline and modified Hough methods.

- Generating an overall comparison of the five available prediction outputs.
- Calculating class-wise mean, median, and maximum angular errors.
- Generating a class-wise angular error heatmap.
- Identifying vehicles with high errors across multiple methods.
- Generating a shared high-error count graph.
- Comparing individual improvements and regressions of Modified Hough.
- Identifying the largest improvement and regression cases.
- Generating visualizations for 19 selected vehicle observations.
- Saving the generated tables, graphs, and visualizations in the Week7_Results folder.

The complete 100-vehicle comparison showed that Modified Hough produced the lowest mean angular error of 5.35° among the evaluated methods.

The shared high-error analysis identified seven vehicle observations with high errors in at least two prediction outputs. The individual improvement analysis showed that Modified Hough reduced the error for 65 vehicles and increased it for 35 vehicles when compared with the Week 6 baseline.

The results provide a better understanding of the performance differences between the methods and establish a basis for the next stage of the project.

6. Work to Be Done Next Week

In the next stage, we plan to focus on the final objective of the project: estimating the complete heading direction of a vehicle, including its front–rear direction.

The work planned includes:

- Reviewing the orientation estimates produced by PCA, Hough Transform, and contour-based methods.
- Preserving the full 0°–180° undirected vehicle axis instead of using only the folded 0°–90° representation.
- Aligning the vehicle crops using the estimated axis.
- Investigating visible and geometric features that may help distinguish the front and rear of vehicles.
- Developing an initial conventional image-processing method for front–rear identification.
- Comparing front–rear predictions with the original dataset heading annotations.
- Evaluating the results and documenting cases where the front and rear cannot be distinguished reliably.

The next stage will extend the axis-estimation work completed during the first seven weeks towards the final objective of estimating the full vehicle heading in the 0°–359° range.

7. References

[1] Y. Bhavsar, M. Zaveri, M. Raval, K. R. Patel, and S. B. Zaveri, “Descriptor: Drone Nadir-view Annotated Images of Vehicles Detection Dataset for India with Heading-angle Oriented Bounding Box (DRASHTI-HaOBB),” *IEEE Data Descriptions*, 2026. Available at: https://ieeexplore.ieee.org/document/11421776.

[2] Y. Bhavsar, M. Zaveri, M. Raval, S. Zaveri, and Ahmedabad University, “DRASHTI-HaOBB: Drone nadir-view annotated images of vehicles dataset for India - heading-angle oriented bounding box,” *Zenodo*, 2026. Available at: https://zenodo.org/records/18278989.